

DECODELABS TASK 02 – BASIC ENCRYPTION AND DECRYPTION

CODE:

```
# Cyber Security Project 2 By Harini Kasi
```

```
# Basic Encryption and Decryption
```

```
# Caesar Cipher
```

```
def encrypt(text, shift):
```

```
    encrypted_text = ""
```

```
    for char in text:
```

```
        if char.isupper():
```

```
            encrypted_text += chr(
                (ord(char) - 65 + shift) % 26 + 65
            )
```

```
        elif char.islower():
```

```
            encrypted_text += chr(
                (ord(char) - 97 + shift) % 26 + 97
            )
```

```
        else:
```

```
            encrypted_text += char
```

```
    return encrypted_text
```

```
def decrypt(text, shift):
```

```
    decrypted_text = ""
```

```
    for char in text:
```

```
        if char.isupper():
```

```
            decrypted_text += chr(
                (ord(char) - 65 - shift) % 26 + 65
            )
```

```
        elif char.islower():
```

```
            decrypted_text += chr(
                (ord(char) - 97 - shift) % 26 + 97
            )
```

```
        else:
```

```

        decrypted_text += char
    return decrypted_text

# User Input
message = input("Enter message: ")
shift = int(input("Enter shift value: "))

# Encryption
encrypted = encrypt(message, shift)
print("\nEncrypted Message:")
print(encrypted)

# Decryption
decrypted = decrypt(encrypted, shift)
print("\nDecrypted Message:")
print(decrypted)

```

OUTPUT

```

PS D:\DecodeLabs> python -u "d:\DecodeLabs\Task2_caesar_cipher.py"
Enter message: Secret
Enter shift value: 3

Encrypted Message:
Vhfuhw

Decrypted Message:
Secret
PS D:\DecodeLabs> python -u "d:\DecodeLabs\Task2_caesar_cipher.py"
Enter message: This is highly confidential
Enter shift value: 2

Encrypted Message:
Vjku ku jkina eqphkfgpvkc n

Decrypted Message:
This is highly confidential
PS D:\DecodeLabs> 

```